KPU CHAMBERS

Law Chambers of Sivaramakrishnan M.S.

Bengaluru

AI PRACTICE OPERATING SYSTEM

Concept Design Document

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1. Executive Summary

This document sets out the concept design for an AI-integrated Practice Operating System (POS) for KPU Chambers, a boutique commercial litigation and advisory practice based in Bengaluru. The system is designed to function as the primary operating layer of the practice — managing matter state, surfacing the right information to the right person at the right time, and executing discrete AI-driven tasks across the full lifecycle of a matter.

The design is grounded in three principles established during the concept development process:

- The system acts on information. Humans act on people.
- Nothing leaves the chambers without passing through a human hand.
- Automation handles the mechanical and administrative. Judgment handles everything that carries the practice's name.

The system is not a replacement for lawyer judgment. It is an infrastructure layer that eliminates administrative friction, surfaces decision-relevant information, and executes well-defined drafting tasks — freeing the legal team to focus on strategy, research, and advocacy.

2. Practice Context

2.1 The Practice

KPU Chambers is a boutique commercial litigation and advisory practice founded and led by Sivaramakrishnan M.S. (Senior Advocate). The practice covers commercial litigation, IBC/insolvency proceedings, arbitration, and corporate advisory work, with a strong emphasis on AI-integrated legal workflows. The Founder holds a Certificate with Honours in AI & Law from Lund University (2025) and is an active thought leader on AI governance and Indian legal profession regulation.

The team currently comprises:

- Sivaramakrishnan M.S. — Founder and Senior Advocate (strategy, arguments, client relationships)
- Vinay Kumar P. — Senior Associate (matter execution, drafting, research)
- Gyana Jha — Associate (matter execution, drafting, research, client updates)

2.2 Current Technology Stack

The practice currently uses the following tools, all of which will be retained and integrated with the new system:

- ECourts (eCourts app) — Cause list tracking, order sheet access, case status for litigation matters
- Microsoft OneDrive — Shared document storage for the office
- Microsoft Outlook — All client and professional communications
- Microsoft Teams — Virtual client meetings and internal coordination

2.3 The Gap the System Addresses

The practice currently operates two matter tracks — litigation and non-litigation — with fundamentally different rhythms. Litigation matters are externally paced by court calendars and are reasonably well tracked through ECourts. Non-litigation matters are internally paced with no mechanical tracking system — they exist in email threads and the Founder's memory.

Across both tracks, the following gaps exist today:

- No single source of matter state — who owns what, where it stands, what is overdue
- The preliminary strategy — the most valuable intellectual output of the intake process — is captured only orally and partially in the engagement letter
- Invoice milestone identification is manual and relies on individual memory
- Client communication after hearings is manual and inconsistent
- There is no Monday morning view of the week's work, priorities, and deadlines

3. System Overview

3.1 What the System Is

The AI Practice Operating System (POS) is the primary operating layer of KPU Chambers. It is a matter-centric system — meaning every document, deadline, communication, task, and invoice hangs off a matter record, and the matter record is the single source of truth for everything happening in the practice.

The system performs three categories of function:

- State management — maintaining an accurate, real-time picture of where every matter stands across both tracks
- AI-driven task execution — performing discrete drafting and analysis tasks using AI capabilities matched to each task type
- Human-in-the-loop orchestration — surfacing the right information and prompts to the right person at the right time, and holding firm checkpoints at every point of professional risk

3.2 What the System Is Not

The system is not an autonomous agent. It does not make decisions, send communications, dispatch invoices, or take actions in the world without human authorisation. It is a preparation and orchestration engine. Everything that leaves the chambers — every email, every document, every invoice — leaves through a human hand from a human device.

Importantly, the system has no Send button. It has a Copy button and a Download button. This is a deliberate design statement about accountability.

3.3 Relationship to Existing Tools

The system sits above the existing technology stack without replacing it:

- ECourts — Remains the authoritative source for order sheets and cause list data. The system consumes this data; it does not replace ECourts.
- OneDrive — Remains the document store. The system holds matter records and links to documents stored in OneDrive; documents do not move.
- Outlook — All client communication originates from Outlook. The system drafts; Outlook sends.
- Teams — VC scheduling remains in Teams. The system may prompt that a call needs to be scheduled; the link is generated in Teams as today.

4. Matter Tracks

The system treats litigation and non-litigation as two distinct tracks with different operating logic, not as variants of a single workflow.

4.1 Litigation Track

Litigation matters are externally paced by court calendars. The ECourts system is the authoritative source for hearing dates and order sheets, with the following qualification on upload timing:

- Supreme Court and Civil / Commercial Courts — Same-day order sheet uploads. System can pull automatically.
- High Court and NCLT — Uploads delayed by approximately one week. System falls back to manual entry by Gyana or the Founder.

The manual fallback is a first-class feature of the system, not an afterthought. Entering a hearing outcome manually must feel as natural as the automated pull.

4.2 Non-Litigation Track

Non-litigation matters are internally paced with no external forcing function. This track currently has the weakest tracking infrastructure — matters exist primarily in email and memory. The POS delivers the highest immediate value on this track by imposing structure where none currently exists.

Non-litigation matters encompass contract drafting, legal opinions, compliance advisory, corporate structuring, and hybrid matters (e.g., consent decrees filed for settlement purposes).

5. Matter Data Model

Every matter in the system is represented as a structured record. The following fields constitute the matter record for each track.

5.1 Litigation Matter Record

Identity

- Matter name (typically Claimant v. Respondent)
- Client name and contact details
- Opposing party name
- Court and jurisdiction
- Case number (populated after filing)
- Type of matter (suit / writ / arbitration / IBC / etc.)

People

- Assigned associate
- Date assigned
- Senior Advocate briefed (if applicable)
- Client contact person (if institutional client)

Timeline

- Date matter opened
- Date engagement letter signed
- Date documents received
- Date of filing
- Next hearing date
- Current stage (pre-filing / filed / arguments / post-order / closed)

Strategy

- Preliminary strategy note (one to two paragraphs — the oral discussion captured in writing)
- Key legal issues identified
- Material deviations noted and resolution recorded

Hearings Log

- Date of each hearing
- Nature of hearing (adjournment / partial arguments / order / etc.)
- Order sheet (attached from ECourts or manually entered)
- Attendee from chambers
- Next date set

Arguments Bundles

- Arguments Bundle Version 1, 2 ... N — each bundle is a paired set of: (a) List of Citations and (b) Synopsis / Notes on Arguments
- Reason for each revision logged (new point by court / new submission by opposing side)
- Final filed bundle marked as such — transition from working document to filed document is a deliberate human action

Fees

- Total engagement value
- Stage-wise fee structure
- Stages invoiced and invoice references
- Payments received and outstanding amount

Documents

- Engagement letter (signed copy)
- Client documents received (index, not the documents — documents reside in OneDrive)
- Drafts sent to client (with version and date)
- Filed documents

5.2 Non-Litigation Matter Record

Identity

- Matter name
- Client name and contact details
- Type of matter (contract drafting / legal opinion / compliance / corporate advisory / etc.)

People

- Assigned associate (if any — some matters are handled directly by the Founder)
- Client contact person

Timeline

- Date matter opened
- Date engagement letter signed
- Date documents received
- Expected delivery date
- Date delivered
- Current stage (pre-engagement / documents received / drafting / client review / delivered / closed)

Strategy

- Approach note (one to two paragraphs)
- Key legal issues identified

Legal Research Note

- Internal research note — not delivered to client but drives the deliverable
- Treated as a working document on the matter record

Action Items

- Open items with owner and deadline

Fees

- Total engagement value
- Stage-wise fee structure (typically Stage 1 on acceptance, Stage 2 on delivery)

- Stages invoiced and invoice references
- Payments received and outstanding amount

Documents

- Engagement letter (signed copy)
- Client documents received (index)
- Drafts sent to client (with version and date)
- Final delivered document

6. End-to-End Workflow

The following is the complete workflow of a matter through the practice, mapped across both tracks with the system's role at each stage.

6.1 Phase 1 — Intake

- Client approaches with an issue via email or note.
- The Founder reads the email, creates a new matter record in the system, and assigns it to an associate.
- The system extracts structured information from the client email (client name, opposing party, nature of dispute, relief sought, urgency indicators, jurisdiction) and pre-populates the matter record for the Founder's confirmation.
- The associate receives a dashboard notification showing the new matter, the client email, and the Founder's deadline for preliminary strategy.

6.2 Phase 2 — Preliminary Strategy

- The Founder reads the client email and conducts preliminary legal research. In parallel, the assigned associate formulates their own preliminary strategy.
- An internal discussion is held. The agreed strategy is captured in the system as the Matter Strategy Note — converting what was previously oral into a written record.
- The Founder schedules a client call or VC (in Teams). The system logs the scheduled call and updates matter status.
- After the client gives the go-ahead, the system prompts preparation of the engagement letter.

6.3 Phase 3 — Engagement

- The engagement letter is generated by the system and reviewed and approved by the Founder before going to the client via Outlook.
- On receipt of the signed engagement letter, matter status updates and the system prompts the document request to the client.
- For litigation matters, the first invoice milestone is flagged on the Founder's dashboard at this point.

6.4 Phase 4 — Document Review

- Client documents are received and indexed on the matter record (documents stored in OneDrive).
- The associate leads the document review manually, comparing documents against the strategy note. The system assists only on specific, targeted queries — not bulk document ingestion.
- If no material deviation: matter proceeds. First invoice milestone confirmed if not already actioned.
- If material deviation: the associate flags it in the system, the Founder is notified, and matter status moves to Strategy Review — Material Deviation. All other triggers are held pending resolution.
- Resolution paths: (a) deviation absorbed — proceed on original track; (b) fee revision required — client communication prompted; (c) new strategy required — new engagement letter flow triggered, old engagement letter terminated.

6.5 Phase 5 — Research and Drafting

- Second round legal research is conducted manually by the associate using SCC Online. The system prompts the task and holds a slot in the matter record for the research note upload.
- The system generates a first draft of the primary deliverable (suit, petition, contract, legal opinion, etc.) for the associate to refine and the Founder to approve.
- Draft is sent to client via Outlook. The system logs the version and date. A follow-up reminder is set if no client response is received within a defined period.
- For non-litigation matters, delivery of the final document and raising of the final invoice typically concludes the matter at this point.

6.6 Phase 6 — Filing (Litigation)

- The associate marks the matter as ready for filing. The Founder is notified.
- On confirmation of filing, the case number is entered into the matter record and matter status moves to Filed.
- Stage 2 invoice milestone is flagged on the Founder's dashboard.

6.7 Phase 7 — Hearings (Litigation)

- Each hearing date triggers a morning reminder to the assigned associate (and to the Founder if the matter is at arguments stage).
- After the hearing: order sheet is pulled from ECourts (for SC and Civil / Commercial Courts) or manually entered (for HC and NCLT). The system generates a draft client update email for the associate to review, copy, and send from Outlook.
- The next hearing date is updated in the matter record. The hearings log entry is created.

6.8 Phase 8 — Arguments (Litigation)

- When the Founder identifies from the dashboard that a matter is listed for arguments, he initiates preparation of the Arguments Bundle.
- Arguments Bundle preparation is conducted manually using SCC Online. The system prompts the task and holds a slot for the bundle upload. The Founder uploads the List of Citations and Synopsis as a paired bundle (Version 1).
- If arguments are adjourned: the reason is logged in the matter record (new point by court / new submission by opposing side). The system prompts preparation of the revised bundle. The previous version is preserved.
- When arguments are concluded, the Founder marks the matter as Arguments Concluded. The system prompts preparation of the filed Synopsis and final List of Citations from the last working bundle.
- The filed versions are uploaded and marked as filed documents. Stage 3 invoice milestone is flagged.

6.9 Phase 9 — Order and Closure

- On receipt of the final order, the Founder is notified. The system prompts client advice on next steps.
- The Founder marks the client as advised. If the matter is fully concluded, status moves to Closed. If further proceedings arise, a new linked matter is opened.
- Final invoice milestone is flagged. Outstanding invoices are surfaced if any remain unpaid.

7. Event and Trigger Logic

The system responds to events — things that happen in the world or actions taken by the team — by updating matter state and surfacing the appropriate next action. The following table maps key events to their triggers.

Event	Trigger / System Response
New matter created and assigned	Matter record created. Associate dashboard updated with matter details and strategy deadline. Founder dashboard shows matter as Pending Internal Strategy Discussion.
Associate submits preliminary strategy note	Founder notified. Dashboard flags matter as Internal Discussion Pending.
Founder approves strategy note	Strategy note locked into matter record. System prompts: schedule client call.
Client go-ahead received	Matter status: Engagement Letter Pending. System prompts: draft engagement letter.
Signed engagement letter uploaded	Matter status: Documents Pending. System prompts: request documents from client. First invoice milestone flagged (litigation).

Event	Trigger / System Response
Client documents received	Associate dashboard: task to review documents against strategy note. Matter status: Document Review.
Document review complete — no deviation	Matter status: Research and Drafting. First invoice milestone confirmed if not already actioned.
Material deviation flagged	Founder notified immediately. Matter status: Strategy Review — Material Deviation. All triggers held pending resolution.
First draft sent to client	Matter record logs draft version and date. Follow-up reminder set. Matter status: Awaiting Client Review.
Client returns draft with comments	Associate dashboard: task to incorporate comments. Matter status: Drafting — Revision.
Final document delivered (non-litigation)	Matter status: Delivered. Final invoice milestone flagged.
Matter marked ready for filing	Founder notified. On filing confirmation: case number entered, matter status: Filed. Stage 2 invoice milestone flagged.
Hearing date arrives	Morning reminder to assigned associate. If arguments stage: reminder to Founder also.
Hearing occurs — order sheet available	Draft client update email generated for associate review. Next date updated in matter record. Hearings log entry created.
Arguments listed — Founder initiates preparation	Task on Founder dashboard: prepare Arguments Bundle Version 1. Matter status: Arguments — Preparation.
Arguments adjourned	Reason logged. Task on Founder dashboard: prepare Arguments Bundle Version N. Previous version preserved.
Arguments concluded	System prompts: prepare filed Synopsis and Citations. On upload of filed versions: Stage 3 invoice milestone flagged. Matter status: Awaiting Order.
Final order received	Founder notified. System prompts: advise client on next steps.
Client advised on next steps	If concluded: matter status Closed. If further proceedings: new linked matter opened. Final invoice milestone flagged. Outstanding invoices surfaced.
Invoice milestone flagged	Milestone appears on Founder's dashboard. Founder initiates invoice generation. Invoice sits in Draft state — cannot be dispatched from this state. Founder approves invoice. Draft email generated. Founder copies and sends from Outlook.

CRITICAL DESIGN RULE	Invoice generation and invoice dispatch are two deliberately separated actions. There is no single-click path from milestone flag to client receipt. The Founder initiates generation, approves the draft, and manually copies and sends from Outlook. The system has no dispatch capability.

8. AI Skills Map

A skill is a discrete AI-driven capability that takes defined inputs, performs a specific task, and produces a defined output. The following twelve skills constitute the AI capability layer of the system.

8.1 Skills Summary Table

Skill	AI Role	Human Role	Confidence
1. Intake and Record Creation	Full AI with human confirmation	Founder confirms and assigns	High
2. Preliminary Legal Research	AI drafts scaffold, human verifies citations	Associate verifies and supplements	Medium-High
3. Strategy Note Generation	AI drafts from structured input	Founder approves and locks	High
4. Engagement Letter Drafting	Full AI generation	Founder reviews and approves	High
5. Document Review and Deviation Analysis	Associate-led; AI on targeted queries only	Associate owns, escalates to Founder	Medium (targeted)
6. Second Round Legal Research	Prompt and reminder only	Associate executes on SCC Online	N/A (human-led)
7. Document Drafting	AI first draft	Associate refines, Founder approves	Medium
8. Client Update Generation	Full AI draft from order sheet	Associate reviews, copies, sends	High
9. Arguments Bundle Preparation	Prompt and reminder only	Founder executes on SCC Online	N/A (human-led)
10. Filed Synopsis Finalisation	AI formatting and structure	Founder reviews before filing	High
11. Invoice Generation	Full AI generation on Founder initiation	Founder initiates, approves, dispatches	High
12. Matter Dashboard and State Management	AI state management and orchestration	Informs only — never acts autonomously	High

8.2 Skill Detail Notes

Skills 6 and 9 — Human-Led Research

The second round of legal research (Skill 6) and the arguments bundle preparation (Skill 9) are deliberately excluded from AI execution. SCC Online is the authoritative source for Indian case law. AI citation generation at the depth required for arguments and filed documents carries an unacceptable hallucination risk. The preliminary research scaffold from Skill 2 provides a starting framework; all substantive research from that point is conducted manually by the team on SCC Online.

Skill 5 — Targeted AI Assistance Only

Document review is associate-led. The volume and complexity of client documents varies significantly across matters. Bulk document ingestion into an AI model creates two compounding risks: context window limitations on large document sets, and higher hallucination probability on complex factual analysis. The associate reads the documents manually, identifies specific discrepancies, and uses AI assistance only on targeted, bounded queries about those specific discrepancies.

The Citation Risk — Applicable to Skills 2, 7, and 8

Across all skills involving legal research or legal content generation, citation accuracy is the primary AI risk. The system is designed to produce research scaffolds and drafts, not authoritative legal documents. Every citation generated by the AI must be verified by a qualified lawyer before reliance. This is a non-negotiable design requirement, not a preference.

9. Dashboard Design

The Monday morning experience is the primary design target for the dashboard. The system should surface the right information to the right person without requiring them to go looking for it.

9.1 Founder Dashboard — Monday Morning View

- New matters added since last login — with client email attached and associate assigned
- Matters where preliminary strategy is pending (internal discussion not yet completed)
- Action items owned by the Founder for the week — across all matters
- All active matters — status, next date, assigned associate (even where no Founder action is required)
- Matters coming up for arguments — with prompt to begin preparation if not already started
- Invoice milestones due — flagged for Founder initiation
- Draft invoices in approval state — awaiting Founder sign-off

9.2 Associate Dashboard — Monday Morning View

- New matters assigned since last login — with client email and strategy deadline from Founder
- Hearings this week — court, date, stage of case
- Action items owned by the associate for the week — across all matters
- Pending tasks with deadlines — document review, drafting, client updates
- Matters where client response is overdue — follow-up prompts

9.3 Client-Facing Communication

Clients currently receive updates only when a hearing has occurred or when the practice needs something from them. The absence of proactive updates generates anxiety and results in inbound calls that interrupt the team's work. A lightweight, periodic matter status communication is identified as a future feature — not designed in Phase 1 — but the system architecture should accommodate it. Any such communication, when implemented, will be associate-reviewed before dispatch from Outlook.

10. Governing Design Principles

The following principles govern every design decision in this system. They are not preferences — they are constraints.

Principle 1	The system acts on information. Humans act on people. Every interaction with a client — every email, every document, every invoice — passes through a human hand before it leaves the chambers.
Principle 2	Nothing leaves the chambers without passing through a human hand. Ever. The system has no Send button. It has a Copy button and a Download button.
Principle 3	Automation handles the mechanical and administrative. Judgment handles everything that carries the practice's name. The four skills carrying the highest legal risk (document review, second-round research, arguments bundle, citation verification) are either human-led or human-verified without exception.
Principle 4	Invoice generation and invoice dispatch are always two separate, deliberate actions. The path from milestone flag to client receipt requires at minimum three conscious human steps. There is no single-click path.
Principle 5	The matter is the organising unit. Every document, deadline, task, communication, and invoice is subordinate to a matter record. The matter record is the single source of truth.
Principle 6	AI confidence must match task risk. High-confidence, low-risk tasks may be more fully automated. Medium-confidence or high-risk tasks require mandatory human review. The four medium-confidence skills (preliminary research, document review, document drafting, arguments bundle) all carry explicit human checkpoints.

11. Future Integrations (Phase 2 and Beyond)

The following integrations are identified as desirable future additions. They are excluded from Phase 1 to maintain architectural clarity and avoid building around moving targets.

- Microsoft Teams VC scheduling — Via Microsoft Graph API, a Teams meeting link could be generated directly from the system when a client call is scheduled. Requires Microsoft 365 tenant configuration for third-party API access.
- Outlook integration — If a reliable and professionally appropriate AI-to-Outlook integration becomes available, the system could generate and stage emails for one-click dispatch from within the system. Currently excluded on accountability grounds and due to the absence of a reliable integration.
- Client-facing portal — A lightweight status view for clients to see where their matter stands without calling the office. Would eliminate a class of inbound interruptions. Requires careful design to avoid over-disclosure of privileged strategy information.
- SCC Online API integration — If SCC Online offers an API, citation retrieval for the preliminary research scaffold (Skill 2) could be automated with authoritative sourcing. Currently SCC Online research is manual for all substantive work.
- ECourts API — More granular integration with ECourts data beyond order sheet retrieval — cause list monitoring, batch hearing alerts across all active matters.

12. Open Questions for Technical Review

The following questions are identified for review by technical advisors:

- Model selection — Which AI model(s) are best suited to each skill category? The preliminary research skill, the drafting skill, and the client update skill may warrant different models optimised for different task types. This requires empirical testing, not theoretical assessment.
- Document ingestion architecture — For Skill 5 (targeted document queries), what is the appropriate architecture for controlled, bounded document ingestion? RAG (Retrieval Augmented Generation) approaches may be appropriate for managing context window constraints on large document sets.
- ECourts data pull — What is the technical mechanism for pulling order sheet data from ECourts? Is there a public API, or does this require a scraping approach? What are the reliability and legal implications of each approach?
- OneDrive integration — What is the appropriate integration model between the POS matter record and OneDrive document storage? The system holds an index of documents; OneDrive holds the documents. How are links maintained when documents are moved or renamed?
- Data residency and confidentiality — Given the privileged and confidential nature of all matter data, what data residency and security architecture is required? This is a Bar Council compliance question as much as a technical one.
- Build approach — Custom build versus configuring an existing practice management platform with AI capabilities layered on top. The custom build approach offers maximum control and specificity; the platform approach offers faster deployment and lower

maintenance burden. This decision should be made with technical input before any build begins.

14. Technical Architecture Layer

This section sets out the proposed technical architecture for the KPU Chambers AI-POS. It was developed through independent AI technical review of the concept design in Sections 1 to 13 and cross-validated across multiple AI systems. It is presented here as the recommended starting point for the engineering team, not as a locked specification.

Note on Status	The concept design in Sections 1 to 13 is functionally confirmed. The technical architecture in this section is a proposed implementation path. Engineering decisions — particularly orchestration framework selection and database choice — remain open for the build team to confirm.

14.1 Infrastructure Layer

The recommended infrastructure stack is Azure-native, chosen for three reasons: it integrates directly with the Microsoft 365 tools already in use (Teams, Outlook, OneDrive); it provides India data residency options compliant with the DPDP Act and advocate-client privilege requirements; and it offers managed AI services that reduce operational complexity for a small practice.

- Identity Provider — Microsoft Entra ID, managed via the KPU Chambers Microsoft 365 tenant. Single sign-on for all team members across the POS and existing Microsoft tools.
- Primary Document Storage — Microsoft OneDrive for Business with India Data Residency. Documents remain in OneDrive. The POS holds matter record metadata and document index links, not the documents themselves.
- AI Compute — Azure OpenAI Service in the South India region (Chennai). GPT-4o for complex drafting and reasoning tasks. GPT-4o-mini for lighter tasks such as classification, extraction, and client update generation. Model selection per skill to be confirmed through empirical testing.
- Vector Database — Azure AI Search in the Central India region (Pune). Powers the RAG architecture for Skill 5 targeted document queries. Documents are chunked, embedded, and stored as vectors. Queries retrieve only relevant chunks rather than ingesting entire documents, solving the context window and hallucination risk identified in the concept design.
- Matter Record Database — Azure Cosmos DB or Azure SQL Database, to be confirmed by the build team. This is the structured database holding all matter records as defined in Section 5. It is distinct from the vector database which holds document content for retrieval.

14.2 OneDrive Integration — Webhook Architecture

The system does not scan OneDrive continuously. It uses a webhook notification model. When a file is added or changed, OneDrive rings a notification bell and the system responds only to that specific event. No polling, no unnecessary scanning, no wasted compute.

Technical implementation: Microsoft Graph Change Notifications with the delta function. When an associate uploads a document to a matter folder, OneDrive sends a change notification to the POS. The system uses the delta function to pull only the specific file that changed, not to re-index the entire directory.

- Step 1 — Associate drops a document into the matter folder on OneDrive
- Step 2 — OneDrive sends a change notification to the POS via Microsoft Graph
- Step 3 — POS pulls only that specific file using the delta function
- Step 4 — File passes through the document processing pipeline: OCR if required, text extraction, chunking, embedding
- Step 5 — Document indexed in Azure AI Search and linked to the matter record by matter ID
- Step 6 — Matter record updated: document received, indexed, available for targeted queries

14.3 Document Processing Pipeline

All documents entering the system pass through the same processing pipeline before they are available for AI-assisted tasks.

- Ingestion — Python-based microservice using Microsoft Graph API handles file synchronisation from OneDrive via the webhook events described above.
- OCR and Text Extraction — Azure Document Intelligence processes PDF and DOCX files into clean text. This covers scanned documents and court orders which may not be machine-readable in their original form.
- Chunking — Text is split using recursive character splitting into overlapping chunks. Chunk size to be calibrated by the build team based on document types common to the practice. Court orders, contracts, and legal opinions have different structural characteristics requiring separate calibration.
- Embedding — Chunks are converted to vector embeddings using the Azure OpenAI text-embedding-3-small model.
- Storage — Embeddings stored in Azure AI Search, linked to the matter record by matter ID.

14.4 Orchestration Layer

The orchestration layer is the traffic controller that sits above all individual skills. It receives a trigger from the matter state engine, determines which skill to invoke, assembles the correct inputs from the matter record and vector store, calls the AI model, processes the output, and routes it back to the matter record and the appropriate dashboard.

Recommended framework: LangGraph or equivalent Azure-native orchestration tooling. This decision is explicitly left open for the build team. The choice of orchestration framework is one of the most consequential early build decisions and should be made by whoever will maintain the system, based on familiarity with the framework and its integration profile with the Azure stack.

Regardless of framework choice, the orchestration layer must be capable of the following:

- Stateful workflow management — tracking where each matter is in its lifecycle and what the next required action is
- Conditional branching — for example, when a material deviation is detected: hold all triggers and route immediately to the Founder
- Skill invocation — calling the correct AI skill with inputs assembled from the matter record and vector store
- Human-in-the-loop gates — holding workflow at approval states until a specific human action is recorded in the database
- Audit logging — recording every AI call, every input provided, every output generated, and every human action taken on that output

14.5 Security and Governance

Security architecture is a first-class design requirement, not a retrofit. The following requirements are non-negotiable.

- Zero Data Retention — All API calls to Azure OpenAI must be configured with ZDR headers. No prompt content, no document content, and no matter data is retained by the model provider for training or any other purpose. This is a requirement under advocate-client privilege and the DPDP Act.
- Approval State Gate — The human-in-the-loop principle is implemented as a mandatory Approval State field in the matter record database. No AI-generated draft can move to Download state without an explicit toggle by an authorised team member. This is the technical implementation of the no Send button principle established in the concept design.
- Audit Log — Every AI invocation is logged locally within the POS: the skill called, the inputs provided, the model used, the output generated, and the human action taken on that output. This log is the evidence trail for the Human Generated AI Assisted declaration and provides defensibility if any AI-assisted output is ever challenged.
- Role-Based Access Control — Microsoft Entra ID manages all access. The Founder has full access across all matters. Associates have access only to matters assigned to them. No team member can access matter records outside their assignment without explicit Founder authorisation.
- Data Residency — All data at rest and in transit must remain within India. Azure South India (Chennai) and Central India (Pune) regions satisfy this requirement. No data may transit through or be stored in non-India Azure regions under any circumstances.

14.6 ECourts Integration

ECourts integration operates in two tiers reflecting the upload latency differences identified in Section 4.1.

- Supreme Court and Civil and Commercial Courts — Same-day order sheet uploads. Automated pull recommended via a third-party legal data aggregator API such as Casemine or equivalent. Important: third-party aggregators are not the authoritative source and introduce their own update latency and licensing costs. The build team must first investigate whether the eCourts National Informatics Centre provides any direct API access. The aggregator route is the fallback, not the default.
- High Court and NCLT — Order sheet uploads delayed by approximately one week. Manual entry by the assigned associate. The manual entry interface must be a first-class feature of the POS, designed to be as natural and frictionless as the automated pull. The two entry paths must produce identical outputs.

In both cases the output is identical: a hearing log entry is created on the matter record, the order sheet text is processed through the document pipeline, and a draft client update email is generated for the associate to review, copy, and send from Outlook.

14.7 Outlook Integration

Two Outlook integration points are identified at different levels of implementation complexity and timeline.

- Inbound email monitoring (Phase 1) — Microsoft Graph API can monitor the chambers inbox for incoming client emails and auto-suggest the creation of a new matter record when an email matches intake patterns. This is a read-only integration. The system reads and analyses the email but does not act without explicit human confirmation. The Founder reviews the suggestion and confirms or dismisses it. No matter record is created without Founder action.
- Outbound draft staging (Phase 2) — If and when a professionally appropriate Outlook integration becomes available, AI-generated draft emails could be staged in Outlook drafts for review and dispatch. Currently excluded on accountability grounds as established in the concept design. If implemented in a future phase, the staged draft must still require a conscious human send action from within Outlook. The POS cannot trigger the send under any circumstances.

14.8 Open Technical Decisions for the Build Team

The following decisions are explicitly held open and must be resolved by the engineering team in consultation with the Founder before any build begins.

- Matter record database — Azure Cosmos DB (document database, flexible schema, higher cost) versus Azure SQL Database (relational, structured, lower cost). The matter record schema defined in Section 5 is relatively stable, which may favour SQL. The build team should assess based on expected query patterns and anticipated scale.
- Orchestration framework — LangGraph versus Azure-native alternatives including Azure Logic Apps, Azure Durable Functions, or Semantic Kernel. Selection should be driven by

the build team's existing expertise and the framework's integration profile with the Azure stack.

- Chunk size calibration — Optimal chunk size for the vector embedding pipeline depends on the document types typical to the practice. Court orders, contracts, and legal opinions have meaningfully different structural characteristics. Calibration requires testing on representative document samples before the pipeline goes live.
- ECourts data source — Direct NIC API if available versus third-party aggregator. This investigation must occur before any aggregator licensing commitment is made.
- Model selection per skill — GPT-4o versus GPT-4o-mini allocation across the twelve skills requires empirical quality and cost testing. The cost differential between models is significant at scale. The allocation should be driven by testing, not assumption.

15. Document Note

This document was prepared in two stages. Sections 1 to 13 were developed through a structured concept design session between the Founder of KPU Chambers and an AI design assistant. The workflow mapping, matter data model, event and trigger logic, skills map, and governing principles all reflect the Founder's actual practice and operational judgment. Section 14 was developed through independent AI technical review of the concept design and cross-validated across multiple AI systems before inclusion.

Sections 1 to 13 are functionally confirmed. They represent decisions made by the Founder about how his practice works and how the system should serve it. Section 14 is a proposed technical implementation path and is presented for engineering review. Open technical decisions are identified explicitly in Section 14.8.

The Founder intends to have this document reviewed by advisors with software engineering and AI systems expertise before any build decisions are made.

Human Generated — AI Assisted

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